Teoría de la Decisión: Decision Problems

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3

Decision Problems

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The previous chapter focused on uncertainty, including how to build probabilistic models of uncertainty and use them to make inferences. This chapter focuses on how to make rational decisions based on a probabilistic model and utility function. We will focus on single-step decisions, reserving discussion of sequential decision problems for the next chapter. This chapter begins by introducing the foundations of utility theory and showing how it forms the basis for rational decision making under uncertainty. We will then show how notions of utility theory can be incorporated into the probabilistic graphical models introduced in the previous chapter to form what are called decision networks. Because many important decision problems involve interacting with other agents, we will briefly discuss game-theoretic models.

3.1 Utility Theory

The previous chapter began by discussing the need to compare the degree of belief with respect to two different statements. This chapter requires the ability to compare the degree of desirability of two different outcomes. We state our preferences using the following operators:

- $A \succ B$ if we prefer A over B.
- $A \sim B$ if we are indifferent between A and B.
- $A \succeq B$ if we prefer A over B or are indifferent.

Just as beliefs can be subjective, so can preferences.

In addition to comparing events, our preference operators can be used to compare preferences over uncertain outcomes. A *lottery* is a set of probabilities associated with a set of outcomes. For example, if $S_{1:n}$ is a set of outcomes and $p_{1:n}$ are their associated probabilities, then the lottery involving these outcomes and probabilities is written

$$[S_1 : p_1; \ldots; S_n : p_n]. \tag{3.1}$$

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This section discusses how the existence of a real-valued measure of utility emerges from a set of assumptions about preferences. From this utility function, it is possible to define what it means to make rational decisions under uncertainty.

3.1.1 Constraints on Rational Preferences

Just as we imposed a set of constraints on beliefs, we will impose some constraints on preferences. These constraints are sometimes called the *von Neumann-Morgenstern axioms*, named after John von Neumann and Oskar Morgenstern, who formulated a variation of these axioms in the 1940s.

- *Completeness.* Exactly one of the following hold: $A \succ B$, $B \succ A$, or $A \sim B$.
- *Transitivity.* If $A \succeq B$ and $B \succeq C$, then $A \succeq C$.
- *Continuity.* If $A \succeq C \succeq B$, then there exists a probability p such that $[A : p; B : 1-p] \sim C$.
- *Independence.* If $A \succ B$, then for any C and probability p, $[A : p; C : 1-p] \succeq [B : p; C : 1-p]$.

These are constraints on *rational preferences*. They say nothing about the preferences of actual human beings; in fact, there is strong evidence that humans are not very rational (Section 3.1.7). Our objective in this book is to understand rational decision making from a computational perspective so that we can build useful systems. The possible extension of this theory to understanding human decision making is of only secondary interest.

3.1.2 Utility Functions

Just as constraints on the comparison of plausibility of different statements lead to the existence of a real-valued probability measure, constraints on rational preferences lead to the existence of a real-valued *utility* measure. It follows from our constraints on rational preferences that there exists a real-valued utility function U such that

- $U(A) > U(B)$ if and only if $A \succ B$, and
- $U(A) = U(B)$ if and only if $A \sim B$.

The utility function is unique up to an *affine transformation*. In other words, for any constants $m > 0$ and b, $U'(S) = mU(S) + b$ if and only if the preferences induced by U' are the same as U. Utilities are like temperatures: you can compare temperatures using Kelvin, Celsius, or Fahrenheit, all of which are affine transformations of each other.

It follows from the constraints on rational preferences that the utility of a lottery is given by

$$U([S_1 : p_1; \ldots; S_n : p_n]) = \sum_{i=1}^{n} p_i U(S_i). \tag{3.2}$$

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Suppose we are building a collision avoidance system. The outcome of an encounter of an aircraft is defined by whether the system alerts (A) and whether a collision results (C). Because A and C are binary, there are four possible outcomes. So long as our preferences are rational, we can write our utility function over the space of possible lotteries in terms of four parameters: $U(a^0, c^0)$, $U(a^1, c^0)$, $U(a^0, c^1)$, and $U(a^1, c^1)$. For example,

$$U([a^0, c^0 : 0.5; a^1, c^0 : 0.3; a^0, c^1 : 0.1; a^1, c^1 : 0.1]) \tag{3.3}$$

is equal to

$$0.5U(a^0, c^0) + 0.3U(a^1, c^0) + 0.1U(a^0, c^1) + 0.1U(a^1, c^1). \tag{3.4}$$

If the utility function is bounded, then we can define a *normalized utility function* where the best possible outcome is assigned utility 1 and the worst possible outcome is assigned utility 0. The utility of each of the other outcomes is scaled and translated as necessary.

3.1.3 Maximum Expected Utility Principle

We are interested in the problem of making rational decisions with imperfect knowledge of the state of the world. Suppose we have a probabilistic model $P(s' \mid a, o)$, which represents the probability that the state of the world becomes s' given that we observe o and take action a. We have a utility function $U(s')$ that encodes our preferences over the space of outcomes. Our *expected utility* of taking action a given observation o is given by

$$EU(a \mid o) = \sum_{s'} P(s' \mid a, o) U(s'). \tag{3.5}$$

The *principle of maximum expected utility* says that a rational agent should choose the action that maximizes expected utility

$$a^* = \arg\max_a EU(a \mid o). \tag{3.6}$$

Because we are interested in building rational agents, Equation (3.6) plays a central role in this book.

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3.1.4 Utility Elicitation

In building a decision making or decision support system, it is often helpful to infer the utility function from a human or a group of humans. This approach is called *utility elicitation* or *preference elicitation*. One way to go about doing this is to fix the utility of the worst outcome $S_\perp$ to 0 and the best outcome $S_\top$ to 1. So long as the utilities of the outcomes are bounded, we can translate and scale the utilities without altering our preferences. If we want to determine the utility of outcome S, then we determine the probability p such that $S \sim [S_\top : p; S_\perp : 1-p]$. It follows that $U(S) = p$.

In our collision avoidance example, the best possible event is to not alert and not have a collision, and so we set $U(a^0, c^0) = 1$. The worst possible event is to alert and have a collision, and so we set $U(a^1, c^1) = 0$. We define the lottery $L(p)$ to be $[a^0, c^0 : p; a^1, c^1 : 1-p]$. To determine $U(a^1, c^0)$, we must find the p such that $(a^1, c^0) \sim L(p)$. Similarly, to determine $U(a^0, c^1)$, we find the p such that $(a^0, c^1) \sim L(p)$.

3.1.5 Utility of Money

It may be tempting to use monetary values to infer utility functions. For example, when one is building a decision support system for managing wildfires, it would be natural to define a utility function in terms of the monetary cost incurred by property damage and the monetary cost for deploying fire suppression resources. However, it is well known in economics that the utility of money, in general, is not linear. If there were a linear relationship between utility and money, then decisions should be made in terms of maximizing expected monetary value. Someone who tries to maximize expected monetary value would have no use for insurance because the expected monetary values of insurance policies are generally negative.

We can determine the utility of money using the elicitation process in Section 3.1.4. Of course, different people have different utility functions, but the function generally follows the curve shown in Figure 3.1. For small amounts of money, the curve is roughly linear—\$100 is about twice as good at \$50. For larger amounts of money, the relationship is often treated as logarithmic. The flattening of the curve makes sense; after all, \$1000 is worth less to a billionaire than it is to the average person.

When discussing monetary utility functions, the three terms below are often used. To illustrate, assume A represents being given \$50 and B represents a 50% chance of winning \$100.

- *Risk neutral.* The utility function is linear. There is no preference between \$50 and the 50% chance of winning \$100 ($A \sim B$).
- *Risk seeking.* The utility function is concave up. There is a preference for the 50% chance of winning \$100 ($A \prec B$).

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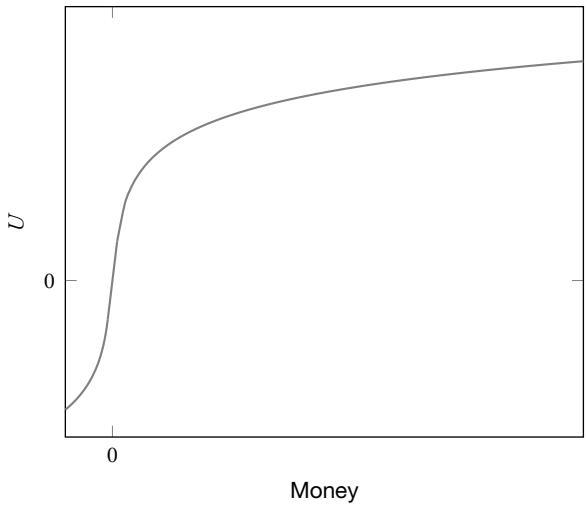

Figure 3.1 Utility of money.

- *Risk averse.* The utility function is concave down. There is a preference for the \$50 ($A \succ B$).

In the process of building decision-making systems, it is useful to use monetary value to inform the construction of the utility function. However, it is important to keep in mind the potentially nonlinear relationship between money and utility.

3.1.6 Multiple Variable Utility Functions

The collision avoidance utility function (Section 3.1.4) depended on two binary variables: whether there was an alert and whether there was a collision. We had to define the utility over all possible combinations of assignments to these two variables. If we had n binary variables, then we would have to specify 2^n parameters in the utility function. If we are able to normalize the utility function, then at least one of the parameters will be 0 and at least one of the parameters will be 1. We can attempt to represent utility functions compactly by leveraging different forms of independence between variables, similar to how Bayesian networks compactly represent joint probability distributions.

Under certain assumptions about the preference structure, we can represent a multiple variable utility function by using a sum of single-variable utility functions. If we have n variables $X_{1:n}$, then we can write

$$U(x_{1:n}) = \sum_{i=1}^{n} U(x_i). \tag{3.7}$$

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To represent the utility function, assuming all variables are binary, we need only $2n$ parameters:

$$U(x_1^0), U(x_1^1), \ldots, U(x_n^0), U(x_n^1). \tag{3.8}$$

To illustrate the value of an additive decomposition of the utility function, we will add two additional variables to our collision avoidance example:

- *Strengthening* (S), which indicates whether the collision avoidance system instructed the pilots to strengthen (or increase) their climb or descent.
- *Reversal* (R), which indicates whether the collision avoidance system instructed the pilots to change direction (i.e., up to down or down to up).

If we did not try to leverage the preference structure over A, C, S, and R, then we would need $2^4 = 16$ parameters. With an additive decomposition, we need only eight parameters.

Although it is common to normalize utilities between 0 and 1, it may be more natural to use a different scheme for constructing the utility function. In the collision avoidance problem, it is easy to formulate the utility function in terms of costs associated with an alert, collision, strengthening, and reversal. If there is no alert, collision, strengthening, or reversal, then there is no cost—in other words, $U(a^0)$, $U(c^0)$, $U(s^0)$, and $U(r^0)$ are all equal to 0. The highest cost outcome is collision, and so we set $U(c^1) = -1$ and normalize the other utilities with respect to this outcome. An alert provides the lowest contribution to the overall cost, and a reversal costs more than a strengthening because it is more disruptive to the pilots. Given our assumptions and keeping $U(c^1)$ fixed at -1, we only have three free parameters to define our utility function.

The utility function for many problems cannot be additively decomposed into utility functions over individual variables as in Equation (3.7). For example, suppose our collision avoidance utility function was defined over three binary variables: whether the intruder comes close horizontally (H), whether the intruder comes close vertically (V), and whether the system alerts (A). A collision threat only really exists if both h^1 and v^1. Therefore, we cannot additively decompose the utility function over the variables independently. However, we can write $U(h, v, a) = U(h, v) + U(a)$.

We can make the additive decomposition explicit in a diagram, as shown in Figure 3.2. The *utility nodes* are indicated by diamonds. The parents of a utility node are *uncertainty nodes* representing the variables on which the utility node depends. If the parents of a utility node are discrete, then the utility function associated with that node can be represented as a table. If the parents of a utility node are continuous, then any real-valued function can be used to represent the utility. If the diagram has multiple utility nodes, their values are summed together to provide an overall utility value.

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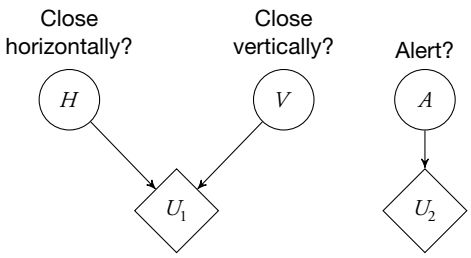

Figure 3.2 Additive decomposition of utility function.

3.1.7 Irrationality

Decision theory is a normative theory that is prescriptive, not a descriptive theory that is predictive of human behavior. Human judgment and preference often do not follow the rules of rationality outlined in Section 3.1.1. Even human experts may have an inconsistent set of preferences, which can be problematic when designing a decision support system that attempts to maximize expected utility.

Tversky and Kahneman studied the preferences of university students who answered questionnaires in a classroom setting. They presented students with questions dealing with the response to an epidemic. The students were to reveal their preference between the following two outcomes:

- A: 100% chance of losing 75 lives
- B: 80% chance of losing 100 lives

Most preferred B over A. From Equation (3.2), we know

$$U(\text{lose } 75) < 0.8U(\text{lose } 100). \tag{3.9}$$

They were then asked to choose between the following two outcomes:

- C: 10% chance of losing 75 lives
- D: 8% chance of losing 100 lives

Most preferred C over D. Hence, $0.1U(\text{lose } 75) > 0.08U(\text{lose } 100)$. We multiply both sides by 10 and get

$$U(\text{lose } 75) > 0.8U(\text{lose } 100). \tag{3.10}$$

Of course, Equations (3.9) and (3.10) result in a contradiction. We have made no assumption about the actual value of $U(\text{lose } 75)$ and $U(\text{lose } 100)$—we did not even assume that losing 100 lives was worse than losing 75 lives. Because Equation (3.2) follows directly from the von Neumann-Morgenstern axioms in Section 3.1.1, there must be a violation of at least one of the axioms—even though many people who select B and C seem to find the axioms agreeable.

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The experiments of Tversky and Kahneman show that certainty often exaggerates losses that are certain relative to losses that are merely probable. They found that this *certainty effect* works with gains as well. A smaller gain that is certain is often preferred over a much greater gain that is only probable, in a way that the axioms of rationality are necessarily violated.

Tversky and Kahneman also demonstrated the *framing effect* using a hypothetical scenario in which an epidemic is expected to kill 600 people. They presented students with the following two outcomes:

- E: 200 people will be saved
- F: $1/3$ chance that 600 people will be saved and $2/3$ chance that no people will be saved

The majority of students chose E over F. They then asked them to choose between:

- G: 400 people will die
- H: $1/3$ chance that nobody will die and $2/3$ chance that 600 people will die

The majority of students chose H over G, even though E is equivalent to G and F is equivalent to H. This inconsistency is due to how the question is framed.

Many other cognitive biases can lead to deviations from what is prescribed by utility theory. Special care must be given when trying to elicit utility functions from human experts to build decision support systems. Although the recommendations of the decision support system may be rational, they may not exactly reflect human preferences in certain situations.

3.2 Decision Networks

We can extend the notion of Bayesian networks introduced in the last chapter to *decision networks* that incorporate actions and utilities. Decision networks are composed of three types of nodes:

- A *chance node* corresponds to a random variable (indicated by a circle).
- A *decision node* corresponds to each decision to be made (indicated by a square).
- A *utility node* corresponds to an additive utility component (indicated by a diamond).

There are three kinds of directed edges:

- A *conditional edge* ends in a chance node and indicates that the uncertainty in that chance node is conditioned on the values of all of its parents.
- An *informational edge* ends in a decision node and indicates that the decision associated with that node is made with knowledge of the values of its parents. (These edges are often drawn with dashed lines and are sometimes omitted from diagrams for simplicity.)

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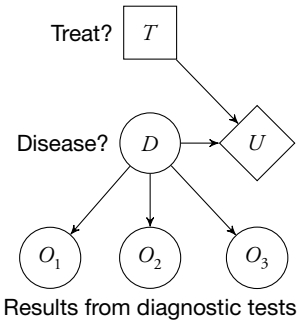

T	D	$U(T,D)$
0	0	0
0	1	−10
1	0	−1
1	1	−1

Figure 3.3 Diagnostic test decision network and utility function.

- A *functional edge* ends in a utility node and indicates that the utility node is determined by the outcomes of its parents.

Decision networks are sometimes called *influence diagrams*. Like Bayesian networks, decision networks cannot have cycles. Representing a decision problem as a decision network allows us to take advantage of the structure of the problem when computing the optimal decision with respect to a utility function. This chapter focuses on *single-shot* decision problems, in which decisions are made simultaneously. The next chapter will focus on problems for which decisions can be made sequentially.

Figure 3.3 shows an example decision network. In this network, we have a set of results from diagnostic tests that may indicate the presence of a particular disease. We need to decide whether to apply a treatment based on what is known about the diagnostic tests. The utility is a function of whether a treatment is applied and whether the disease is actually present. This network is used as a running example in this section.

3.2.1 Evaluating Decision Networks

As introduced in Section 3.1.3, the expected utility of a given o is

$$EU(a \mid o) = \sum_{s'} P(s' \mid a, o) U(s'). \tag{3.11}$$

The s' in Equation (3.11) represents an instantiation of the nodes in the decision network. We can use the equation to compute the expected utility of treating a disease for the decision network in Figure 3.3. For now, we will assume that we have the result from only the first diagnostic test and that it came back positive. If we wanted to make the knowledge of the first diagnostic test explicit in the diagram, then we would draw

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an informational edge from O_1 to T, and we would have

$$EU(t^1 \mid o_1^1) = \sum_{o_3} \sum_{o_2} \sum_{d} P(d, o_2, o_3 \mid t^1, o_1^1) U(t^1, d, o_1^1, o_2, o_3). \tag{3.12}$$

We can use the chain rule for Bayesian networks and the definition of conditional probability to compute $P(d, o_2, o_3 \mid t^1, o_1^1)$. Because the utility node depends only on whether the disease is present and whether we treat it, we can simplify $U(t^1, d, o_1^1, o_2, o_3)$ to $U(t^1, d)$. Hence,

$$EU(t^1 \mid o_1^1) = \sum_{d} P(d \mid t^1, o_1^1) U(t^1, d). \tag{3.13}$$

Any of the exact or approximate inference methods introduced in the previous section can be used to evaluate $P(d \mid t^1, o_1^1)$. To decide whether to apply a treatment, we compute $EU(t^1 \mid o_1^1)$ and $EU(t^0 \mid o_1^1)$ and make the decision that provides the highest expected utility.

To evaluate general single-shot decision networks, we begin by instantiating the action nodes and observed chance nodes. We then apply any inference algorithm to compute the posterior over the parents of the utility nodes. Instead of summing over the instantiation of all variables in Equation (3.11), we only need to sum over the instantiations of the parents of the utility nodes. The optimal decision is the one that, when instantiated in the decision network, provides the highest expected utility.

A variety of methods have been developed over the years to make evaluating decision networks more efficient. One method involves removing action and chance nodes from decision networks if they have no children, as defined by conditional, informational, or functional edges. For example, in Figure 3.3, we can remove O_2 and O_3 because they have no children. We cannot remove O_1 because we treated it as observed, indicating there is an informational edge from O_1 to T (although it is not drawn explicitly).

3.2.2 Value of Information

So far, we have assumed that for the decision network in Figure 3.3, we have only observed o_1^1. Given the positive result from that one diagnostic test alone, then we may decide against treatment. However, it may be beneficial to administer additional diagnostic tests to reduce the risk of not treating a disease that is really present. One way to decide which diagnostic test to administer is to compute the *value of information*.

In computing the value of information, we will use $EU^*(o)$ to denote the expected utility of an optimal action given observation o. The value of information about variable O' given o is

$$VOI(O' \mid o) = \left(\sum_{o'} P(o' \mid o) EU^*(o, o') \right) - EU^*(o). \tag{3.14}$$

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In other words, the value of information about a variable is the increase in expected utility with the observation of that variable. The expected utility can only increase if the observation of the variable can lead to a different optimal decision. If observing a new variable O' makes no difference in the choice of action, then $EU^*(o, o') = EU^*(o)$ for all o', in which case Equation (3.14) evaluates to 0. For example, if the optimal decision is to treat the disease regardless of the outcome of the diagnostic test, then the value of observing the outcome of the test is 0.

The value of information only captures the increase in expected utility from making an observation. There may be a cost associated with making a particular observation. Some diagnostic tests may be quite inexpensive, such as a temperature reading; other diagnostic tests are more costly and invasive, such as a lumbar puncture. The value of information obtained by a lumbar puncture may be much greater than a temperature reading, but the costs of the tests should be taken into consideration.

The value-of-information metric is an important and often used metric for choosing what to observe. Sometimes the value-of-information metric is used to determine an appropriate sequence of observations. After each observation, the value of information is determined for the remaining unobserved variables. The unobserved variable with the greatest value of information is then selected for observation. If there are costs associated with making different observations, then these costs are subtracted from the value of information when determining which variable to observe. The process continues until it is no longer beneficial to observe any more variables. The optimal action is then chosen. This greedy selection of observations is only a heuristic and may not represent the truly optimal sequence of observations. The optimal selection of observations can be determined by using the techniques for sequential decision making introduced in later chapters.

3.2.3 Creating Decision Networks

Decision networks are a powerful framework for building decision support systems. So far, we have discussed the key elements in building decision networks and how to use them to make optimal single-shot decisions. We will briefly discuss the procedure for creating a decision network.

The first step is to identify the space of possible actions. For an airborne collision avoidance system, the actions may be to climb, descend, or do nothing. In some problems, it may be desirable to factor the space of actions into multiple decision variables. In a collision avoidance system that can recommend both horizontal and vertical maneuvers, one decision variable may govern whether we climb or descend and another variable may govern whether we turn left or right.

The next step is identifying the observed and unobserved variables relevant to the problem. If we have an electro-optical sensor on our collision avoidance system, we may

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be able to observe the relative angle to another aircraft, and so this angular measurement would correspond to one of the observed chance nodes. The true position of an intruding aircraft is relevant to the problem, but it cannot be observed directly, and so it is represented by an unobserved variable in the network.

We then identify the relationships between the various chance and decision nodes. Determining these relationships can be done by using expert judgment, learning from data as discussed in Section 2.4, or a combination of both. Often designers try to choose the direction of arrows to reflect causality from one node to another.

Once the relationships of the chance and decision nodes are identified, we choose models to represent the conditional probability distributions. For discrete nodes, an obvious model is a tabular representation. For continuous nodes, we can choose a parametric model such as a linear Gaussian model. The parameters of these models can then be specified by experts or estimated from the data by using the techniques introduced in Section 2.3.

We introduce the utility nodes and add functional edges from the relevant chance and decision nodes. The parameters of the utility nodes can be determined from preference elicitation from human experts (Section 3.1.4). The parameters can also be tuned so that the optimal decision according to the decision network matches the decisions of human experts.

The decision network should then be validated and refined by human experts. Given a decision scenario, the decision network can be used to determine the optimal action. That action can be compared against what a human expert would recommend. Generally, inspection of many decision scenarios is required before confidence can be established in a decision network.

If there is disagreement between the decision network and a human expert, then the decision network can be inspected to determine why that particular action was selected as optimal. In some cases, closer inspection of the model may result in revising the conditional probabilities, modifying the relationship between variables, changing the parameters in the utility nodes, or introducing new variables into to the model. Sometimes further study results in the human experts revising their choice of action. Several development iterations may be required before an appropriate decision network is found.

3.3 Games

This chapter has focused on making rational decisions with an assumed model of the environment. The methods introduced in this chapter so far can, of course, be applied to environments containing other agents so long as our probabilistic model captures the effects of the behavior of the other agents. However, there are many cases in which we do not have a probabilistic model of the behavior of the other agents, but we do have a

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		Agent 2	
		Testify	Refuse
Agent 1	Testify	$-5, -5$	$0, -10$
	Refuse	$-10, 0$	$-1, -1$

Figure 3.4 Prisoner's dilemma.

model of their utilities. Making decisions in such situations is the subject of *game theory*, which will be briefly discussed in this section.

3.3.1 Dominant Strategy Equilibrium

One of the most famous problems in game theory is the *prisoner's dilemma*. We have two agents who are prisoners and are being interrogated separately. Each is given the option to testify against the other. If one testifies and the other does not, then the one who testifies gets released, and the other gets ten years in prison. If both testify, then they both get five years. If both refuse to testify, then they both get one year.

The utilities of the two agents in the prisoner's dilemma are shown in Figure 3.4. The first component in the utility matrix is associated with Agent 1, and the second component is associated with Agent 2. It is assumed that the utility matrix is common knowledge between the two agents. The two agents must select their actions simultaneously without knowledge of the other agent's action.

In games like the prisoner's dilemma, the strategy chosen by an agent in a game can be either a

- *pure strategy*, in which an action is chosen deterministically, or
- *mixed strategy*, in which an action is chosen probabilistically.

Of course a pure strategy is just a special case of a mixed strategy in which probability 1 is assigned to a single action. The mixed strategy that assigns probability 0.7 to testifying in the prisoner's dilemma can be written using a notation similar to that used earlier for lotteries:

$$[\text{Testify} : 0.7; \text{Refuse} : 0.3].$$

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The utility of a mixed strategy can be written in terms of utilities associated with pure strategies:

$$U([a_1 : p_1; \ldots; a_n : p_n]) = \sum_{i=1}^{n} p_i U(a_i). \tag{3.15}$$

The strategy of agent i is denoted s_i. A *strategy profile*, $s_{1:n}$, is an assignment of strategies to all n agents. The strategy profile of all agents except for that of agent i is written s_{-i}. The utility of agent i given a strategy profile $s_{1:n}$ is written $U_i(s_{1:n})$ or, alternatively, $U_i(s_i, s_{-i})$.

A *best response* of agent i to the strategy profile s_{-i} is a strategy s_i^* such that $U_i(s_i^*, s_{-i}) \geq U_i(s_i, s_{-i})$ for all strategies s_i. There may be, in general, multiple different best responses given s_{-i}. In some games, there may exist an s_i that is a best response to all possible s_{-i}, in which case s_i is called a *dominant strategy*. For example, in the prisoner's dilemma, Agent 1 is better off testifying regardless of whether Agent 2 testifies or refuses. Hence, testifying is the dominant strategy for Agent 1. Because the game is symmetric, testifying is also the dominant strategy for Agent 2. When all agents have a dominant strategy, their combination is called a *dominant strategy equilibrium*.

The prisoner's dilemma has generated significant interest because it shows how playing individual best responses can result in a suboptimal outcome for all agents. The dominant strategy equilibrium results in both agents testifying and receiving five years in prison. However, if they both refused to testify, then they would have both only received one year in prison.

3.3.2 Nash Equilibrium

Suppose we have two aircraft on a collision course, and the pilots of each aircraft must choose between climb or descend to avoid collision. If the pilots both choose the same maneuver, then there is a crash with utility -4 to both pilots. Because climbing requires more fuel than descending, there is an additional penalty of -1 to any pilot who decides to climb. The utility matrix is shown in Figure 3.5.

In the collision avoidance game, there does not exist a dominant strategy equilibrium. The best response of a particular pilot depends on the decision of the other pilot. There is an alternative equilibrium concept known as the *Nash equilibrium*. A strategy profile is a Nash equilibrium if no agent can benefit by switching strategy, given that the other agents adhere to the profile. In other words, $s_{1:n}$ is a Nash equilibrium if s_i is a best response to s_{-i} for all agents i.

There are two pure-strategy Nash equilibria in the collision avoidance game, namely, (Climb, Descend) and (Descend, Climb). It has been proven that every game has at least one Nash equilibrium, which may or may not include pure strategies. There are no known polynomial time algorithms for finding Nash equilibria for general games,

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		Agent 2	
		Climb	Descend
Agent 1	Climb	$-5, -5$	$-1, 0$
	Descend	$0, -1$	$-4, -4$

Figure 3.5 Collision avoidance game.

although the complexity of finding Nash equilibria is not NP-complete (instead it belongs to the complexity class known as PPAD).

3.3.3 Behavioral Game Theory

When one is building a decision-making system that must interact with humans, information about the Nash equilibrium is not always helpful. Humans often do not play a Nash equilibrium strategy. First of all, it may be unclear which equilibrium to adopt if there are many different equilibria in the game. For games with only one equilibrium, it may be difficult for a human to compute the Nash equilibrium because of cognitive limitations. Even if human agents can compute the Nash equilibrium, they may doubt that their opponents can perform that computation.

An area known as *behavioral game theory* aims to model human agents. Many different behavioral models exist, but the *logit level-k model*, sometimes called the *quantal level-k model*, has become popular recently and tends to work well in practice. The logit level-k model captures the assumption that humans are

- more likely to make errors when those errors are less costly, and
- limited in the number of steps of strategic look-ahead (as in "I think that you think that I think...").

The model is defined by

- a *precision parameter* $\lambda \geq 0$, which controls sensitivity to utility differences (0 is insensitive); and
- a *depth parameter* $k > 0$, which controls the depth of rationality.

In the logit level-k model, a level-0 agent selects actions uniformly. A level-1 agent assumes the opponents adopt level-0 strategies and selects actions according to the logit distribution

$$P(a_i) \propto e^{\lambda U_i(a_i, s_{-i})}, \tag{3.16}$$

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where s_{-i} represents the assumed strategy profile of the other agents. A level-k agent assumes that the other agents adopt level $k-1$ strategies and select their own actions according to Equation (3.16). The parameters k and λ can be learned from data by using techniques discussed in the previous chapter.

To illustrate the logit level-k model, we will use the *traveler's dilemma*. In this game, an airline loses two identical suitcases from two travelers. The airline asks the travelers to write down the value of their suitcases, which can be between \$2 and \$100, inclusive. If both put down the same value, then they both get that value. The traveler with the lower value gets their value plus \$2. The traveler with the higher value gets the lower value minus \$2. In other words, the utility function is as follows:

$$U_i(a_i, a_{-i}) = \begin{cases} a_i & \text{if } a_i = a_{-i} \\ a_i + 2 & \text{if } a_i < a_{-i} \\ a_{-i} - 2 & \text{otherwise} \end{cases}. \tag{3.17}$$

Most people tend to put down between \$97 and \$100. However, somewhat counterintuitively, there is a unique Nash equilibrium of only \$2.

Figure 3.6 shows the strategies of the logit level-k model with different values for λ and k. The level-0 agent selects actions uniformly. The level-1 agent is peaked toward the upper end of the spectrum, with the precision parameter controlling the spread. As k increases, the difference between the strategies for $\lambda = 0.3$ and $\lambda = 0.5$ becomes less apparent. Human behavior can often be modeled well by logit level 2; as we can see, this provides a better model of human behavior than the Nash equilibrium.

3.4 Summary

- Rational decision making combines probability and utility theory.
- The existence of utility function follows from constraints on rational preferences.
- A rational decision is one that maximizes expected utility.
- We can build rational decision-making systems based on utility functions inferred from humans.
- Humans are not always rational.
- Decision networks compactly represent decision problems.
- Behavioral game theory is useful when making decisions involving multiple agents.

3.5 Further Reading

The theory of expected utility was initiated by Bernoulli in 1793 [2]. The axioms for rational decision making presented in Section 3.1.1 are based on those of Neumann and Morgenstern in their classic text, *Theory of Games and Economic Behavior* [3]. Neumann

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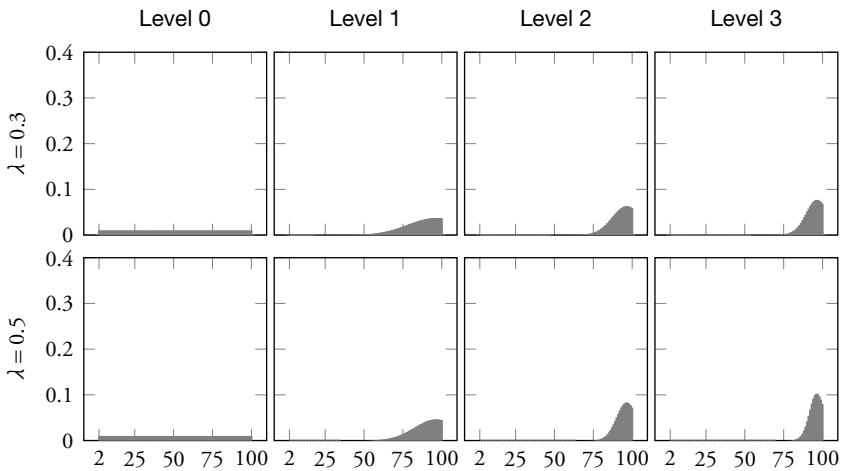

Figure 3.6 Logit level-k for traveler's dilemma.

and Morgenstern prove that those axioms lead to the existence of a utility function and establish the basis for the maximum expected utility principle [3]. Schoemaker provides an overview of the development of utility theory [4], and Fishburn surveys the field [5]. Russell and Norvig discuss the importance of the maximum expected utility principle to the field of artificial intelligence [6].

Farquhar surveys a variety of methods for utility elicitation [7], and Markowitz discusses the utility of money [8]. *Decisions with Multiple Objectives: Preferences and Value Tradeoffs* by Keeney and Raiffa provides an overview of multiple attribute utility theory [9]. The book discusses the assumptions about preference structure that allow certain kinds of utility function decompositions, including the additive decomposition discussed in Section 3.1.6.

The example of irrational preferences in Section 3.1.7 is taken from Tversky and Kahneman [1]. Kahneman and Tversky provide a critique of expected utility theory and introduce an alternative model called prospect theory that appears to be more consistent with human behavior [10]. Several recent books discuss human irrationality, including *Predictably Irrational: The Hidden Forces That Shape Our Decisions* [11] and *How We Decide* [12].

The textbook *Bayesian Networks and Decision Graphs* by Jensen and Nielsen discusses decision networks. Early papers by Shachter provide algorithms for evaluating decision networks [14], [15]. Howard introduces the quantitative concept of the value of information [16], which has been applied to decision networks [17], [18].

Game theory is a broad field, and there are several standard introductory books [19]–[21]. Koller and Milch extend decision networks to a game-theoretic context [22].

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